WQF7003
Intelligent Computation

Logic

What is logic?

- The study of reasoning
- Focuses on the relationship among statements as opposed to the content of any particular statement.
- Instead of focusing on the statements, we use symbols to represent them.

>> Symbolic logic

common advertisement logic fallacies

common advertisement logic fallacies

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Common advertisement logic fallacies include the bandwagon appeal, which suggests that a product is good because many people use it, and the appeal to authority, where a celebrity endorses a product without relevant expertise. Other examples are false dilemmas, which present limited choices, and ad hominem attacks that discredit competitors instead of focusing on the product's merits.

W Wikipedia clutch.co

More

Auto-generated based on listed sources. May contain inaccuracies. Was this helpful?

AS audiencescience.com
https://www.audiencescience.com › ads-with-logical-fallacies

Ads With Logical Fallacies: Types & Real-Life Examples

Jun 30, 2025 · What Is Fallacy Advertising? Commercials using **fallacies** deliver a misleading statement about the product or service to trigger audiences' emotional connection and cognitive biases, grabbing their attention and persuading them to take immediate action (making a purchase). Logical **fallacies**, in...

common politics logic fallacies

common politics logic fallacies

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2 Go2Tutors
https://go2tutors.com › 16-logical-fallacies-politicians-use-daily

16 Logical Fallacies Politicians Use Daily - go2tutors.com

Jul 15, 2025 · **Politics** is full of fast talk, grand gestures, and clever comebacks. But underneath all the noise, there's often some sneaky **logic** being used to shape what people think. These tricks don't always show up as lies, but they can twist the truth just enough to make things sound better—or worse—than...

LinkedIn
https://www.linkedin.com › pulse › unmasking-manipulation-top-29-fallacies-political-rhetoric-jan...

Unmasking Manipulation: The Top 29 Fallacies in Political ... - Link...

Nov 22, 2024 · In today's highly polarized world, political speeches often disguise misleading arguments through logical **fallacies**. Learning to identify these **fallacies** is essential for critical thinking and ...

josricardomartins.substack.com
https://josricardomartins.substack.com › p › unmasking-political-rhetoric-a-logical

Unmasking Political Rhetoric: A Logical Guide to Detecting False ...

Mar 21, 2025 · I offer this practical guide to identifying logical inconsistencies, starting with an overview of verbal **logic**, the distinction between inductive and deductive reasoning, and the most **common fallacies** and sophisms used in political discourse and TV commentators.

Types of logic

- Propositional logic
- First order logic
- Probabilistic Logic
- Fuzzy Logic
- …

Statement

- Statement/Proposition : a sentence that is either TRUE or FALSE, but not both.
  - I can run.
  - I cannot run
  - 2 > 1
  - 1 > 2
- We can use variables (a, b, c, P, Q, R, …) to represent propositions.

Logical connectives

- NOT
- AND
- OR
- IF … THAN
- IFF

Negation, ¬

- Negation of p, denoted $\neg p$, is the proposition:

  not p

- Example:

  p : Bicycles have windscreens

  $\neg p$: Bicycles do not have windscreens

- Or, in English, we can use "It is not the case that p" to express a negation.
- Notations: $\neg p$, $\sim p$, $!p$, p' or $\overline{p}$.

AND, $\wedge$

- We can combine propositions with AND.
- Example:

  p: 2+2=4 q: birds can swim.
- "p $\wedge$ q" is "2+2=4 and birds can swim."

Truth table

P	Q	P $\wedge$ Q
T	T	T
T	F	F
F	T	F
F	F	F

OR, $\vee$

- We can combine propositions with OR.
- Example:

  p: 2+2=4 q: birds can swim.
- "p $\vee$ q" is "2+2=4 or birds can swim."

Truth table

P	Q	P $\vee$ Q
T	T	T
T	F	T
F	T	T
F	F	F

Conditional Proposition

- The propositions in the form

  if p then q

- are conditional propositions that denote as :

  $p \rightarrow q$.

- Example:

  If you are in Kuala Lumpur,

  then you are in Malaysia.

- p: you are in KL (antecedent)
- q: you are in Malaysia (consequent)

Truth Table of Conditional Proposition

Given: If you are in KL, than you are in Malaysia

p	q	$p \rightarrow q$	
T	T	T	You are in KL, you are in Malaysia
T	F	F	
F	T	T	
F	F	T	

Truth Table of Conditional Proposition

Given: If you are in KL, than you are in Malaysia

p	q	$p \rightarrow q$	
T	T	T	You are in KL, you are in Malaysia
T	F	F	You are in KL, but not in Malaysia? No way!
F	T	T	
F	F	T	

Truth Table of Conditional Proposition

Given: If you are in KL, than you are in Malaysia

p	q	$p \rightarrow q$	
T	T	T	You are in KL, you are in Malaysia
T	F	F	You are in KL, but not in Malaysia? No way!
F	T	T	You are in Sabah!
F	F	T	You are in Sabah!

Do not violate the definition in the given $p \rightarrow q$, or, not falsified by the given condition

Biconditional Proposition

- Bidirectional Propositions – in the form:

  p if and only if q

- Example : Integer x is divisible by 2 if and only if x is an even number.
- Denote as : $p \leftrightarrow q$
- Can be written as p iff q
- Equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$

Truth table

p	q	$p \rightarrow q$	$q \rightarrow p$	$p \rightarrow q \wedge q \rightarrow p$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

Order of Precedence

Operator	***Precedence***
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

Logical Equivalent

- Assume A and B are two propositions,

  A = … … …

  B = … …

- A and B are "logically equivalent", or

  $A \leftrightarrow B$

- iff A and B have the same truth values.
- 2 ways to determine logical equivalent:
  - Truth Table
  - Laws of Logic

Truth Table

- 2 propositions are logical equivalent if they have same truth values
- Example: Prove Law of Implication [show that p → q and ~pVq are logically equivalent]

TABLE 4 Truth Tables for $\neg p \vee q$ and $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Same truth values
→ logical equivalent

Laws of Logic

$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws

Laws of Logic

$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

$p \rightarrow q \equiv \neg p \vee q$ Law of Implication

Example

Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology

$$(p \wedge q) \rightarrow (p \vee q)$$

$$= \neg(p \wedge q) \vee (p \vee q)$$ Implication law

$$= (\neg p \vee \neg q) \vee (p \vee q)$$ De Morgan's law

$$= \neg p \vee \neg q \vee p \vee q$$ Associative law

$$= (\neg p \vee p) \vee (\neg q \vee q)$$ Associative law

$$= T \vee T = T$$ Negation law

Deductive Reasoning

Deductive Reasoning

- Consider the following sequence of propositions.

a) We are looking for a man either in black or yellow shirt
b) The man with black shirt is wearing jeans.
c) The man we are looking for is not wearing jeans

- Assuming that these statements are true, it is reasonable to conclude:

The man we are looking for is wearing yellow shirt

- drawing a conclusion from a sequence of propositions → Deductive Reasoning

Deductive Reasoning

The following is an **argument**.

a) We are looking for a man either in black or yellow shirt
b) The man with black shirt is wearing jeans.
c) The man we are looking for is not wearing jeans

Therefore, the man we are looking for is wearing yellow shirt

(a), (b), (c) are called **hypothesis / premises**

"Therefore, … …." is the **conclusion**

Valid or Invalid

- In a valid argument, we say that the conclusion follows from the hypotheses.
- Note: We are not saying that the conclusion is true; we are only saying that if you grant the hypotheses, you must also grant the conclusion.
- An argument is valid because of its form, not because of its content.

Rules of Inference

- Rules of inference are methods for deriving conclusions from premises.
- They form a core component of formal logic, providing the structure that makes arguments valid.
- When an argument with true premises adheres to a rule of inference, its conclusion must also be true.

Rules of Inference

Modus Ponens

$p \rightarrow q$

p

$\therefore q$

Rules of Inference

Modus Ponens

$$p \rightarrow q$$
$$p$$
$$\text{---------}$$
$$\therefore q$$

Example

If this object is made of copper, it will conduct electricity.

This object is made of copper.

Therefore, it will conduct electricity

Rules of Inference

Modus Tollens

$p \rightarrow q$

$\sim q$

$\therefore \sim p$

Rules of Inference

Modus Tollens

$p \rightarrow q$

$\sim q$

$\therefore \sim p$

Example

If I am hungry I eat a lot

I don't eat a lot

So I am not hungry

Rules of Inference

Hypothetical Syllogism

$$p \rightarrow q$$
$$q \rightarrow r$$

$$\therefore p \rightarrow r$$

Rules of Inference

Hypothetical Syllogism

$$p \rightarrow q$$
$$q \rightarrow r$$

$$\therefore p \rightarrow r$$

Example

If it rains, we will not have a picnic.

If we don't have a picnic, we won't need a picnic basket.

Therefore, if it rains, we won't need a picnic basket.

Rules of Inference

Dilemma

$$p \vee q$$
$$p \rightarrow r$$
$$q \rightarrow s$$

$$\therefore r \vee s$$

Rules of Inference

Dilemma

$$p \vee q$$
$$p \rightarrow r$$
$$q \rightarrow s$$
$$\text{----------}$$
$$\therefore r \vee s$$

Example

Either we take LRT or we drive

If we take LRT, we pay the fare

If we drive, we pay tolls

So, either we pay LRT fare or we pay tolls

There are some common fallacy

<u>Affirming the consequent</u>

$p \rightarrow q$	If am in Thailand, then I am in Asia.
q	I am in Asia.
---	Therefore I am in Thailand.
$\therefore p$	

There are some common fallacy

✗ <u>Denying the antecedent</u>

$p \rightarrow q$	If am in Thailand, then I am in Asia.
$\sim p$	I am not in Thailand.

$\therefore \sim q$	Therefore I am not in Asia.

Example

- Identify the conclusion is valid or not:

  “If birds can fly or swim, then they can find food. Birds can fly. Tigers can find food. Therefore, birds and tigers can find food.”

First Order Logic / Predicate Logic

Statements that no one can tell true or false

- x is greater than 10.
- There is an old man who can cure cancer without using any medicine.

x is greater than 10.

Subject Predicate (refers to a property that the subject may have)

These statements can be denoted by **propositional function,** e.g. G(x) or C(x)

Quantifiers

- Quantification : to create a proposition from a propositional function.
- In English, the words <u>all</u>, <u>some</u>, <u>many</u>, <u>none</u>, and <u>few</u> are used in quantification.
- We focus on 2 types of quantification: universal and existential quantification.

Universal Quantification

- assert that **a property is true for all values** of a variable in the **domain of discourse**.
  - i.e. P(x) is true for all values of x in this domain.
- The domain must always be specified when a universal quantifier is used, since a change of domain may lead to change of truth value or unverifiable.

Universal Quantification

- The universal quantification of P(x) is the statement:
- “P(x) for all values of x in the domain.”
- Notation:

$$\forall x\ P(x)$$

- ∀ is universal quantifier. Read as “for all”, “for every”, “for each”, ...

Existential Quantification

- Proposition that is true if and only if P(x) is true for at least one value of x in the domain.
- The existential quantification of P(x) is the proposition:
- “There exists an element x in the domain such that P(x).”
- Notation : $\exists x P(x)$

Existential Quantification

- $\exists$ is called the existential quantifier, read as “there exists,” “for some,” “for at least one,” or “there is.”
- $\exists x P(x)$ is read as:

  “There is an x such that P(x),”

  “There is at least one x such that P(x),”

  or

  “For some x P(x).”

Examples

- Let P(x) be the statement “x + 1 > x” for all real numbers x. Is ∀xP(x) true? _____
- Let Q(x) be the statement “x < 2” where the domain consists of all real numbers. Is ∀xQ(x) true? _____
- Let Q(x) be the statement “x < 2” where the domain consists of all real numbers. Is ∃xQ(x) true? _____
- Let R(x) be the statement “x is the sibling of x” where the domain consist of everyone in the world. Is ∃xR(x) true? _____

Domains of Quantifiers

- The domains of quantifiers can be expressed in mathematical form.
- Example:

$\forall x \in \mathbb{R}$, $P(x) = \ldots \ldots$

$\forall x \in \mathbb{R}$, $x+1>x$

$\exists x \in \mathbb{R}$, $(x/2 = x)$

Examples of Quantified Statements

- “If x likes BLACKPINK then he likes K-Pop”
- $B(x) \rightarrow K(x)$
- Proposition?

- “For all people it holds that if the person likes BLACKPINK then she likes K-Pop”
-
- Proposition?

Examples of Quantified Statements

- “If x likes BLACKPINK then he likes K-Pop”
- $B(x) \rightarrow K(x)$
- Proposition? No

- “For all people it holds that if the person likes BLACKPINK then she likes K-Pop”
- $\forall x(B(x) \rightarrow K(x))$
- Proposition? Yes

Examples of Quantified Statements

- “x is your classmate who has founded Microsoft”
- $C(x) \wedge M(x)$
- Proposition?

- “There is a person who is your classmate and the founder of Microsoft”
-
- Proposition?

Examples of Quantified Statements

- “x is your classmate who has founded Microsoft”
- $C(x) \wedge M(x)$
- Proposition? No

- “There is a person who is your classmate and the founder of Microsoft”
- $\exists x\ (\ C(x) \wedge M(x)\ )$
- Proposition? Yes

Probabilistic logic

Probability

- Probability of X happening = (Number of ways X can happen) / (Total number of outcomes)
- Example: Throwing a dice and the outcome is 3 → 1/6
- Example: Toss a coin and get a Head → ½
- Example: Throw dice and toss coin together, and get “3” and “Head”? _____

Probabilistic logic

- combines the capacity of probability theory to handle uncertainty with the formal structure of deductive logic.
- While classical logic deals with 0 and 1, probabilistic logic uses the continuous interval [0,1].
- A value of P(A)=0.7 indicates a 70% degree of likelihood that proposition A is true.

Example

- Instead of:

  If you are tall, you are a good basketball player.

- Probabilistic logic

  If you are tall, you are more likely to be a good basketball player.

  (75%???)

Example

- Instead of:

  If you follow this treatment, you will be cured

- Probabilistic logic

  If you follow this treatment, there is a chance you will be cured

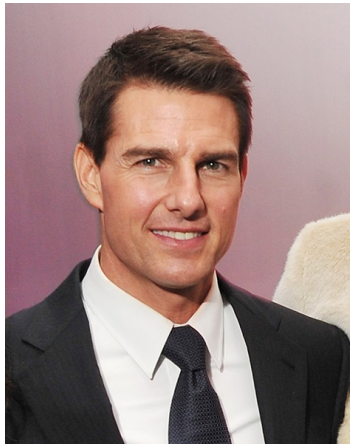
  (1%???? 50%??? 95%???)

Fuzzy Logic

Fuzzy Logic

- Deal with uncertainty and vagueness
- Example: Who is the shortest tall man?

Harry Potter
165cm

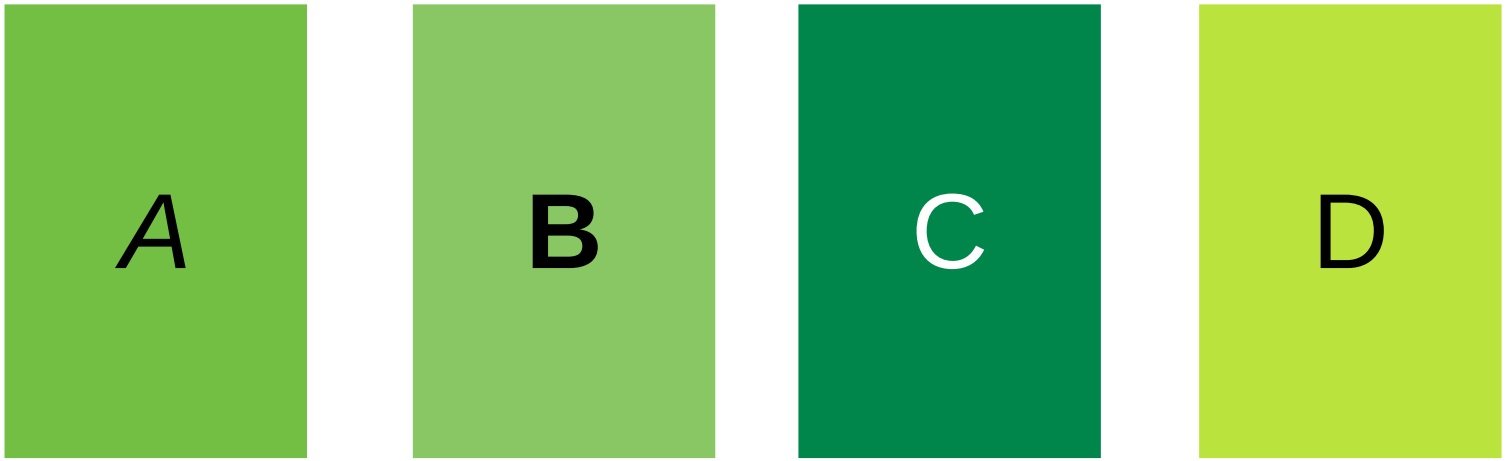


Tom Cruise
172cm

Johnny Depp
175cm

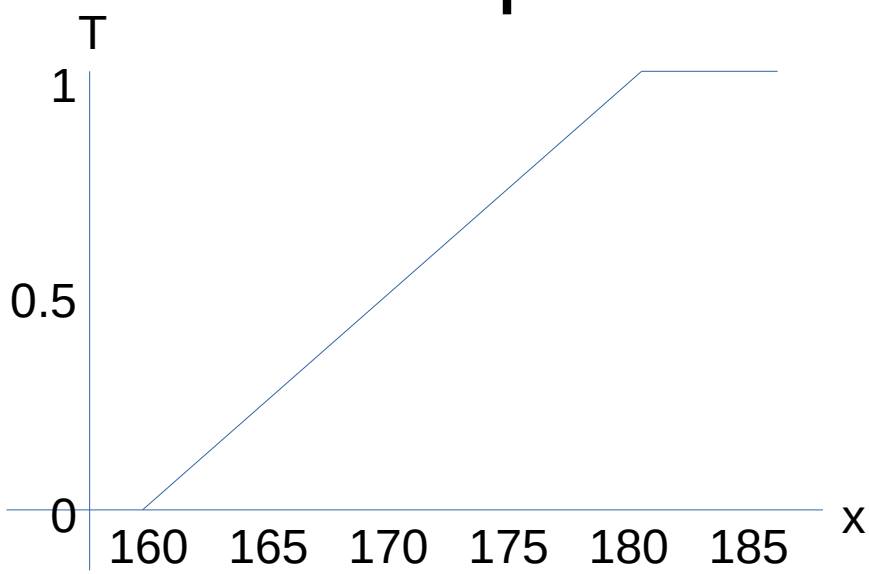
Daniel Craig
178cm

- Or, which one is green?



Fuzzy Sets

- Green, Tall Man, etc are all fuzzy sets.
- Each element in a fuzzy set comes with a membership degree in [0,1]
- Membership degrees are defined by membership functions



Leonardo DiCaprio : 181cm

Johnny Depp : 175cm

Harry Potter : 165cm

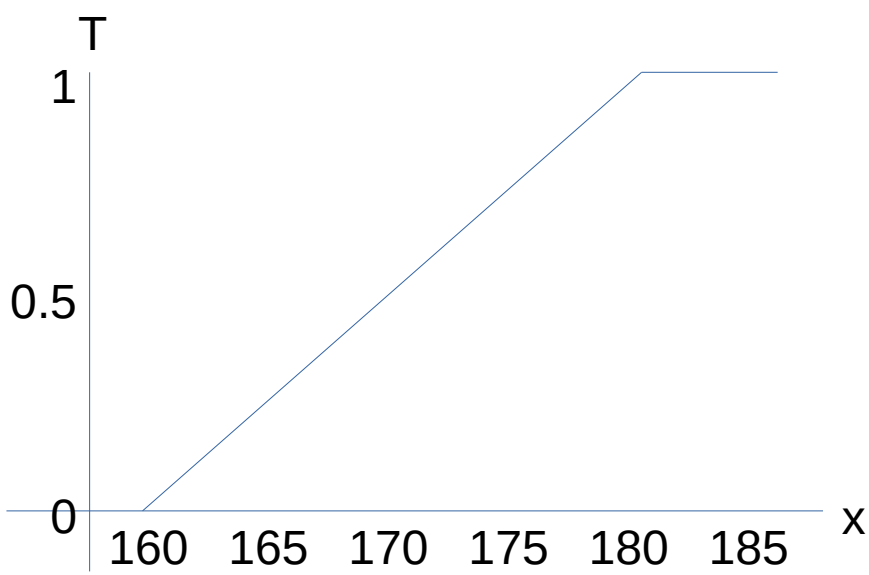
T(Leonardo DiCaprio) = 1.0

T(Johnny Depp) = 0.75

T(Harry Potter) = 0.25

Fuzzy Sets

- T – fuzzy set Tall
- T(x) – fuzzy membership degree for Tall
- Johnny Depp and Harry Potter are elements in fuzzy set Tall
- T(Harry Potter) = 0.25 → Harry Potter's degree in Tall



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T(Leonardo DiCaprio) = 1.0
T(Johnny Depp) = 0.75
T(Harry Potter) = 0.25

Fuzzy rules and fuzzy logic (example)

- If you are tall, you are a good basketball player.
- T(x) : what is the membership degree of x in Tall
- B(y) = what is the membership degree of y as a good basketball player?
- What is the relationship between T and B?

Any question?